# CIT 314 Mobile Application Development

# Practical 1: Setup Flutter Development Environment

**Step 1: Install Git**

1. Download Git from the official Git website.
2. Install Git using the default settings.
3. Open **Git Bash**.
4. Check installation by :

```
git --version
```

**Step 2: Download Flutter SDK**

1. Download the Flutter SDK.
2. Extract it to a suitable location, for example:

```
C:\src\flutter
```

3. Add the Flutter `bin` folder to the Windows **PATH**.

```
C:\src\flutter\bin
```

**Step 3: Verify Flutter Installation**

Open Git Bash or Command Prompt and type:

```
flutter --version
```

Then run:

```
flutter doctor
```

Check the displayed results and identify any missing requirements.

**Step 4: Install Visual Studio Code**

1. Download and install Visual Studio Code.
2. Open VS Code.
3. Install the following extensions:
    - Flutter
    - Dart

**Step 5: Verify Flutter in VS Code**

1. Open VS Code.
2. Open the Command Palette using:

```
Ctrl + Shift + P
```

3. Search for:

```
Flutter: New Project
```

4. Verify that VS Code can detect the Flutter SDK.

# Practical 2: Install and Work with Git

**Step 1: Check Git Installation**

Open **Git Bash** and type:

```
git --version
```

Git version information should be displayed.

**Step 2: Configure Git**

Set your name:

```
git config --global user.name "Your Name"
```

Set your email:

```
git config --global user.email "youremail@edu.com.pk"
```

Verify the configuration:

```
git config --global --list
```

**Step 3: Create LAB3 Folder**

Create a new folder named:

```
LAB3
```

You can create it using Git Bash:

```
mkdir LAB3
```

Move into the folder:

```
cd LAB3
```

**Step 4: Initialize Git Repository**

Initialize the repository:

```
git init
```

Git will create a hidden `.git` folder inside `LAB3`.

**Step 5: Design a Screen in Figma**

1. Open Figma.
2. Create a simple mobile application screen.
3. Save/export the required design file.
4. Place the design file inside the **LAB3** folder.

For example:

```
LAB3
│
└── login-screen.fig
```

**Step 6: Check Repository Status**

Run:

```
git status
```

Git will show the files that are currently untracked or modified.

**Step 7: Add Files to Staging Area**

```
git add .
```

Check the status again:

```
git status
```

**Step 8: Create the First Commit**

```
git commit -m "Add Figma screen design"
```

**Step 9: Check Git Log**

```
git log
```

The commit information should be displayed.

# Practical 3: Working with Git and GitHub

## Step 1: Create Repository on GitHub

1. Open GitHub.
2. Sign in to your GitHub account.
3. Click **New repository**.
4. Enter repository name:
   `LAB3`
5. Keep the repository empty initially.
6. Click **Create repository**.

## Step 2: Connect Local Repository with GitHub

Open Git Bash inside your local `LAB3` folder.
Add the GitHub repository as a remote:

```
git remote add origin https://github.com/yourusername/LAB3.git
```

Replace `yourusername` with your actual GitHub username.
Check the remote:

```
git remote -v
```

## Step 3: Check the Current Branch

```
git branch
```

If your branch is `master`, push using:

```
git push -u origin master
```

If your branch is `main`, use:

```
git push -u origin main
```

**Note:** For new repositories, `main` is commonly used. If Practical 2 repository was initialized with `master`, you can either continue with `master` or rename it to `main`.
To rename `master` to `main`:

```
git branch -M main
```

Then:

```
git push -u origin main
```

## Step 4: Verify Repository on GitHub

1. Open your GitHub repository.
2. Refresh the page.
3. Check that your files and commit are visible.

## Step 5: Clone the Repository

To download a copy of the GitHub repository:

```
git clone https://github.com/yourusername/LAB3.git
```

A new `LAB3` folder will be created.
Move into the cloned repository:

```
cd LAB3
```

Check its status:

```
git status
```